



From powder to product: a perspective on halide electrolytes for commercial lithium solid-state batteries

Nicholas S. Grundish^{1,2}

Received: 30 May 2025 / Revised: 2 September 2025 / Accepted: 8 September 2025
© The Author(s) 2026

Abstract

Halide solid electrolytes have emerged as compelling candidates for next-generation solid-state batteries, offering a promising balance of ionic conductivity, interfacial stability, and processability. Compared to sulfides and oxides, halides have a unique blend of properties—high oxidative stability for high-voltage cathodes, good mechanical deformability, and compatibility with emerging anode interfaces. This perspective examines the state of the field, highlighting structure–property relationships across trigonal, spinel, and emerging oxyhalide frameworks, and emphasizing how aliovalent doping, mixed-anion strategies, and Earth-abundant chemistries expand the halide design space. Challenges in moisture sensitivity, interphase formation, and long-term chemical stability are discussed, while scalable synthesis pathways—from mechanochemical milling to melt processing—and their trade-offs in cost and phase purity are evaluated. Integration strategies for halides into composite electrodes and full-cell architectures are examined, with a focus on manufacturability and performance metrics relevant to commercialization. To contextualize these findings, a comparative analysis with sulfide and oxide systems is provided, along with a roadmap for halide-based solid-state battery development to guide future academic research and industrial development initiatives. Halide solid electrolytes now represent a major class of solid electrolyte materials that may provide the critical mix of properties to push true all-solid-state batteries closer to a commercial reality.

Keywords Tungsten · Solid-state batteries · Lithium batteries · Solid electrolytes · Halide electrolytes

1 Introduction

In the rapidly evolving landscape of solid-state battery (SSB) technology, halide-based solid-state electrolytes (SSEs) have emerged as a compelling class of materials. Historically overshadowed by sulfide and oxide electrolytes, especially in the commercial development landscape, halides—composed of lithium cations and halogen anions (e.g., Cl^- , Br^- , I^-)—have recently garnered attention due to their promising combination of electrochemical performance, processability, and high-voltage stability [1, 2]. This renewed interest is not solely driven by performance benchmarks; rather, it reflects a broader shift in the field toward manufacturability and holistic cell-level design.

Several converging trends have positioned halide solid electrolytes at the forefront of next-generation battery research. These include the development of ambient-pressure synthesis techniques, the rising availability of doped halide frameworks with high ionic conductivities, and a growing awareness of the limitations of sulfide-based systems—particularly their poor oxidative stability and processing hazards [3, 4]. In contrast, halides offer comparatively benign processing environments, reduced interfacial degradation, and chemical compatibility with high-voltage oxide cathodes. Importantly, the moderate mechanical modulus of halides suggests potential for better interface conformity and lower stress buildup than oxides, without the volatility of sulfides.

This perspective aims to explore halide solid electrolytes as an emerging class of materials capable of addressing many longstanding bottlenecks in SSB development. Their unique attributes and unresolved challenges are explored, giving way to new possibilities for academic and industrial research directions needed to translate laboratory insights into scalable, robust battery systems for lithium-based chemistries.

✉ Nicholas S. Grundish
nicholas.grundish@utexas.edu

¹ McKetta Department of Chemical Engineering, University of Texas, Austin 78712, USA

² Energy Exploration Technologies Inc., Austin 78754, USA

2 Material attributes

Halide solid electrolytes occupy a distinct position within the broader spectrum of solid-state ionic conductors, offering performance characteristics that challenge long-held assumptions about trade-offs between ionic conductivity, electrochemical stability, and processability. Select halide compositions now routinely exhibit room-temperature ionic conductivities exceeding $1 \text{ mS}\cdot\text{cm}^{-1}$ —a benchmark often considered the threshold for practical battery implementation [5–7]. This level of performance is especially notable given the structural simplicity of many halide phases, which display this level of ionic conductivity in simple three-element systems. Through aliovalent doping, the performance of these materials can be further improved, as vacancy- or interstitial-mediated diffusion pathways can be created to balance conductivity with material stability. This factor is evidenced through the evaluation of the reversible uptake of moisture for some materials in the $\text{Li}_2\text{M}^{4+}\text{Cl}_6$ – $\text{Li}_3\text{M}^{3+}\text{Cl}_6$ system, while others irreversibly degrade [8].

Equally significant is the broad oxidative stability window demonstrated by halide electrolytes. In contrast to sulfides, which exhibit pronounced instability at voltages above $\sim 3 \text{ V}$ vs. Li/Li^+ , halide systems—particularly chlorides and bromides—maintain electrochemical compatibility with high-voltage layered oxide cathodes, including $\text{Li}[\text{Ni}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}]\text{O}_2$ (NMC811) [1, 9]. This intrinsic stability obviates the need for complex protective coatings or conservative voltage limitations, thereby enabling the design of higher-energy density SSB architectures and providing a more streamlined supply chain from a cathode active material sourcing standpoint.

Mechanically, halides occupy an intermediate regime—softer than oxides but not quite as soft as sulfides, and they are generally more chemically benign than their sulfide counterparts. These material qualities confer several advantages in terms of manufacturability. Many halides can be densified through cold pressing or low-temperature sintering without the application of extreme pressures or exposure to corrosive environments. Additionally, their moderate compatibility with dry organic solvents facilitates slurry-based processing routes, opening a pathway toward integration with established electrode fabrication lines.

Perhaps most compelling is the relatively low grain boundary (GB) resistance observed in halide systems compared to oxide-based conductors [10]. This property reduces the need for high-temperature sintering to achieve interparticle connectivity, thereby enabling the use of fine powders and less energy-intensive shaping techniques. The inherently lower surface reactivity of halides further supports long-term chemical and mechanical stability, minimizing undesirable side reactions during cycling and storage.

To contextualize the comparative advantages of halide electrolytes, Table 1 presents a side-by-side evaluation of key material metrics, including ionic conductivity, electrochemical stability, mechanical modulus, interfacial behavior, and process compatibility—across halide, sulfide, and oxide families. This analysis highlights the “Goldilocks” position that halides may occupy: combining the cathode compatibility and oxidative stability absent in sulfides with the manufacturability and benign processing behavior not typically found in oxides. Realizing this potential at scale, however, will require mitigation of moisture sensitivity and further optimization of interfacial integration strategies.

3 Halide solid electrolyte design landscape

The halide solid electrolyte family offers a chemically diverse and structurally tunable platform for materials innovation. Unlike oxides and sulfides, which often rely on a narrow set of host frameworks, halides accommodate a wide array of cationic substitutions and anionic configurations—enabling a combinatorial design space that remains largely untapped.

One of the most powerful strategies in halide electrolyte development, as with other solid electrolyte systems, is aliovalent doping. Alternative cations can be introduced to tailor the lithium content in the material and either create vacancies or cause interstitial lithium that enhances ionic conductivity. For example, in the $\text{Li}_{2-z}\text{Mg}_{1-3z/2}\text{Zr}_z\text{Cl}_4$ system, partial substitution of Mg^{2+} with Zr^{4+} leads to ionic conductivity improvement for $z \geq 0.4$ by optimizing the balance between mobile carrier concentration and lattice integrity [14]. Importantly, such doping strategies also modify local structural distortions, impact defect chemistry, and tune bulk mechanical properties—making them useful not just for transport optimization but also for cell-level integration. Additionally, substituting more expensive elements with more common elements can lead to a decrease in raw material costs as well as performance improvement [16].

Crystal structure plays a critical role in dictating ion transport pathways and dimensionality. Halides can crystallize into a variety of crystallographic space groups, each offering different diffusion pathways and ion migration mechanisms. Figure 1 illustrates representative crystal structures across four major halide electrolyte families—trigonal (e.g., Li_3YCl_6), monoclinic (e.g., Li_3InCl_6 or Li_2ZrCl_6), spinel type (e.g., Li_2MgCl_4), and the newly emerging oxyhalide class of materials (e.g., LiNbOCl_4)—emphasizing the diversity in anion and cation coordination environments that underpin differences in ionic transport behavior [11, 12, 17]. For lithium-based halide solid electrolytes relevant to room-temperature, commercial applications, the UCl_3 -type

Table 1 Comparative properties of halide, sulfide, and oxide solid electrolytes

Property	Halide (e.g., Li_3YCl_6)	Sulfide (e.g., $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_6\text{PS}_5\text{Cl}$)	Oxide (e.g., $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ —LLZO, $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ —LAGP)	Refs.
Ionic conductivity	$\sim 0.1\text{--}10\text{ mS}\cdot\text{cm}^{-1}$ at 25 °C (composition dependent)	$\sim 1\text{--}20\text{ mS}\cdot\text{cm}^{-1}$ at 25 °C (very high in some compositions)	$\sim 10^{-3}\text{--}1\text{ mS}\cdot\text{cm}^{-1}$ at 25 °C (generally lower)	[11–13]
Electrochemical stability window	$\sim 0.5\text{--}4.5\text{ V}$ vs. Li^+ (wide, especially oxidative side)	$\sim 1.5\text{--}2.5\text{ V}$ vs. Li^+ (limited by oxidative decomposition)	$\sim 0\text{--}6\text{ V}$ vs. Li^+ (very wide, especially vs. cathodes)	[11–14]
Moisture sensitivity	Moderate to high (hydrolysis forms HCl, degradation)	Very high (forms H_2S gas, rapid degradation)	Low (generally air stable, inert to moisture)	[11–13]
Thermal stability	Moderate (stable to $\sim 250\text{--}300\text{ }^\circ\text{C}$)	Low to moderate ($\sim 150\text{--}300\text{ }^\circ\text{C}$ depending on system)	High ($> 500\text{ }^\circ\text{C}$, suitable for co-sintering)	[11]
Mechanical modulus (Young's)	$\sim 10\text{--}30\text{ GPa}$ (soft/moderate)	$\sim 5\text{--}20\text{ GPa}$ (very soft, deformable)	$\sim 100\text{--}200\text{ GPa}$ (rigid, brittle)	[11, 13, 14]
Dendrite suppression	Moderate (intermediate modulus, interface dependent)	Poor (soft, prone to dendrite penetration)	Good (high modulus resists dendrites)	[11, 14]
Processability	Good (cold-pressable, low-temp processing possible)	Excellent (plastic, easily densified)	Poor (requires high-temp sintering, sensitive to defects)	[11, 12, 14]
Processing atmosphere compatibility	Requires a dry room; sensitive to humidity	Requires an inert or dry room; highly reactive in air	Compatible with air (in powder form); sintering may require a controlled environment	[8, 11, 15]
Cathode compatibility	High (chemically stable vs. high-voltage cathodes)	Poor (oxidatively unstable, needs coatings)	High (inert vs. cathodes, but poor interface contact)	[11, 13, 14]
Anode compatibility	Moderate (may form passivating interphases)	Poor (high reactivity with Li metal)	Good (LLZO in particular, if the interface is engineered)	[11, 13, 14]
Interphase stability	Moderate (stable vs. cathodes; passivation vs. Li)	Poor (decomposition at both electrodes)	High (chemically inert, but can suffer from contact loss)	[11, 13, 14]
GB resistance	Low (good ion transport across particles)	Very low (minimal GB impedance)	High (GBs often limit conductivity)	[13, 14]
Synthesis complexity	Moderate (may require dry room, high-purity precursors)	Low to moderate (but toxic gases evolve)	High (complex and high-temp processes)	[11–13]
Toxicity/environmental risk	Moderate (HCl possible during hydrolysis)	High (H_2S evolution in air/moisture)	Low (generally inert)	[8, 11, 13]
Material cost (approx.)	Moderate to high (rare-earth salts like YCl_3 are expensive)	Low to moderate (sulfur abundant, but Ge, P costly)	High (e.g., Ta, Nb, Ga in LLZO variants expensive)	[11, 13, 14]
Research maturity	Emerging (rapidly growing body of work)	Mature (well-studied, multiple commercial efforts)	Mature (especially LLZO and LAGP systems)	

hexagonal structure is not typically observed. This hexagonal UCl_3 -type symmetry is usually encountered in extreme synthetic conditions for lithium-based halide materials or in sodium-based halide systems [18–20]. The structural motifs depicted in Fig. 1 offer distinct ion diffusion topologies and highlight the opportunity to tailor conduction pathways through lattice engineering. Although it is outside the scope of this perspective, the effect of crystal structure and composition on Li^+ transport mechanisms in halide solid electrolyte materials has been reviewed elsewhere [11, 12, 19–23].

Among the most widely studied halide electrolytes is the Li_3YCl_6 family, which serves as a model system for understanding structure–property relationships in trigonal halide frameworks. This material exhibits room-temperature ionic conductivities on the order of $\sim 0.5\text{--}1.5\text{ mS}\cdot\text{cm}^{-1}$, depending on synthesis method and dopant concentration. Its structure

consists of close-packed Cl^- anions with lithium occupying both octahedral and tetrahedral interstitial sites, enabling three-dimensional ion transport pathways [1]. Isovalent doping with trivalent cations such as In^{3+} , Sc^{3+} , or Al^{3+} has been shown to increase carrier concentration by introducing lithium vacancies, often improving conductivity and stabilizing phase formation [7, 24, 25]. Such substitutions can induce crystallization in the monoclinic phase, as observed in Li_3InCl_6 or aliovalent Zr^{4+} -substituted Li_2ZrCl_6 [24, 26]. Both the trigonal and monoclinic polymorphs of the Li_2MCl_6 and Li_3MCl_6 families exhibit relatively high oxidative stability—extending beyond 4.5 V vs. Li^+ —and display chemical compatibility with high-voltage layered oxide cathodes such as NMC811. These characteristics have established them as benchmarking compositions within the halide electrolyte literature. Nonetheless, persistent challenges

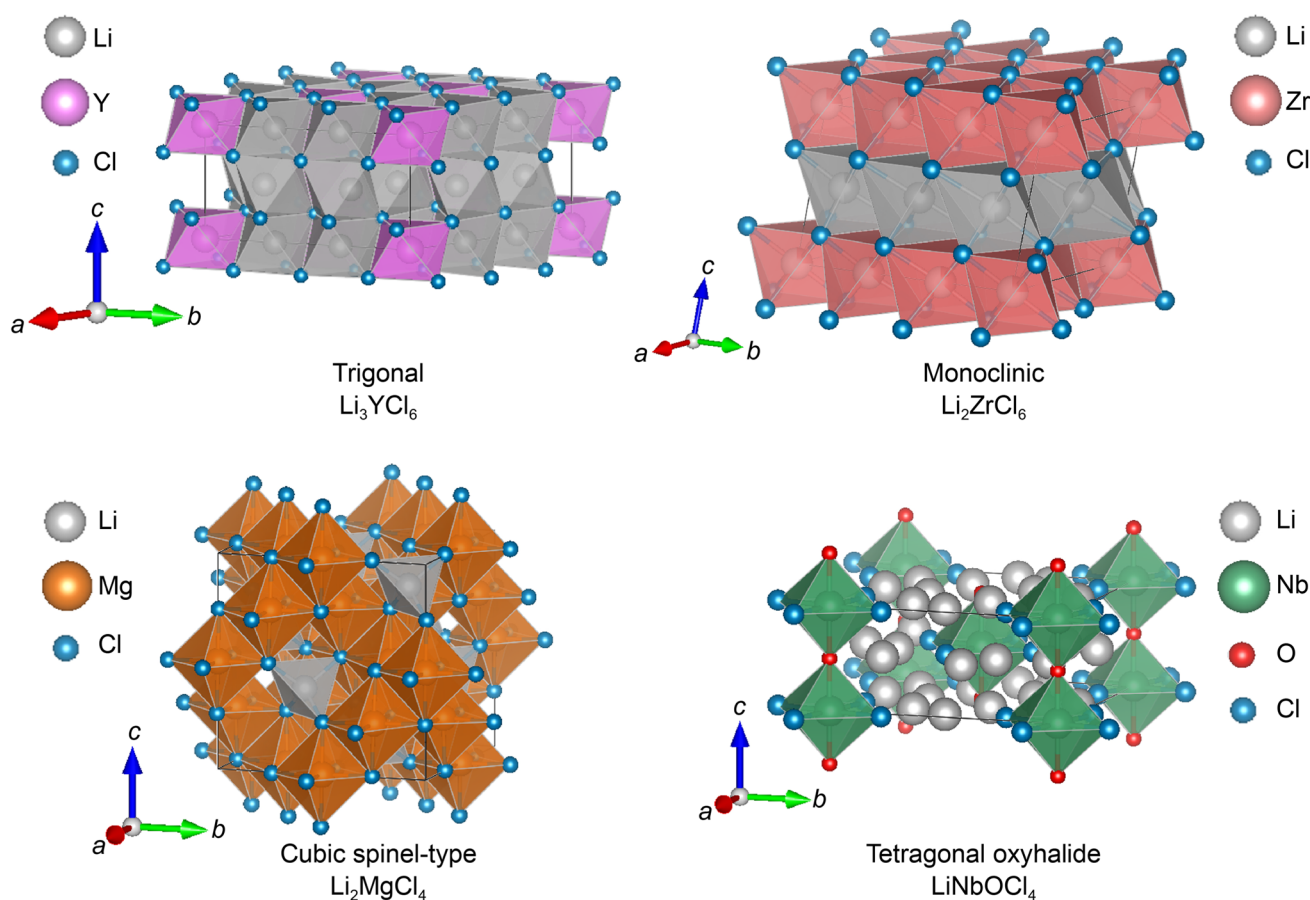


Fig. 1 Representative lithium halide solid electrolyte framework crystal structures

include controlling phase purity, mitigating air sensitivity during synthesis and processing, and ensuring long-term stability during cycling. Despite these limitations, both trigonal and monoclinic members of the Li_2MCl_6 – Li_3MCl_6 family remain central to mechanistic studies and continue to inform the design of next-generation halide chemistries.

Although originally discovered in the early 1980s, recent attention has returned to spinel-type halide electrolytes, which offer a new design archetype [27]. Spinel halides such as Li_2MCl_4 ($\text{M}=\text{M}^{2+}$) feature interconnected tetrahedral and octahedral sites that facilitate 3D ion conduction. These materials show promise not only for their conductivity but also for their structural stability and mechanical integrity—both critical for maintaining interfacial contact in composite electrodes. Li_2MgCl_4 is a model system for this spinel electrolyte framework. In this system, Li^+ occupies the tetrahedral 8a sites while Mg^{2+} resides in octahedral 16d positions, creating interconnected pathways for Li^+ migration. Although the reported ionic conductivity of undoped Li_2MgCl_4 remains modest ($\sim 10^{-6}$ – 10^{-7} $\text{S}\cdot\text{cm}^{-1}$ at room temperature), the spinel structure offers a tunable platform

for aliovalent or isovalent doping strategies to enhance vacancy concentration and mobility [27]. By doping additional Mg^{2+} or Zn^{2+} on the lithium sites in Li_2MgCl_4 , the ionic conductivity can increase several orders of magnitude to 10^{-4} – 10^{-5} $\text{S}\cdot\text{cm}^{-1}$ [14, 27, 28]. Alternative spinel systems demonstrate ionic conductivity on the order of 10^{-3} $\text{S}\cdot\text{cm}^{-1}$, as in the case of $\text{Li}_2\text{Sc}_{2/3}\text{Cl}_4$ [29]. Additionally, spinels tend to exhibit greater thermodynamic stability and improved tolerance to mechanical stress compared to layered or trigonal analogs, which may benefit interfacial integrity during long-term cycling. The relatively simple binary composition and use of abundant elements also support favorable cost and scalability profiles. However, detailed studies on Li^+ transport mechanisms, moisture sensitivity, and phase stability in spinel halides are still limited, making this class a fertile target for future exploration and optimization.

Mixed-anion systems, particularly oxyhalides, represent another promising frontier. LiNbOCl_4 is a representative example of this emerging class of solid electrolytes, which integrates halide and oxide anions within a single lattice to exploit the complementary benefits of each. There

is currently some debate as to the exact structure of these materials, as they were originally reported to be orthorhombic, but recently, the X-ray diffraction pattern has been shown to be better fitted with a tetragonal space group [30]. For demonstrative purposes, the tetragonal version of this material is portrayed in Fig. 1. In this structural depiction, Nb^{5+} centers are coordinated by a mixed array of O^{2-} and Cl^- anions, forming isolated NbOCl_4 polyhedra with a disordered Li^+ sublattice filling in the space between them. This structure provides ample pathways for Li^+ conduction through the material, as this class of materials has examples (LiNbOCl_4 and LiTaOCl_4) with some of the highest reported lithium conductivities among the halide materials and surpassing the ionic conductivities of conventional liquid electrolytes, making it one of the fastest Li^+ conducting solids reported in the literature [5]. The partial substitution of chlorides with oxides may also reduce the hygroscopic nature typical of pure halide systems, improving air and moisture stability, but this aspect of these materials has yet to be experimentally validated. This system exemplifies how mixed-anion design can unlock novel structural motifs and property combinations not accessible in more traditional halide or oxide frameworks alone. Aluminum oxyhalides have recently emerged as a distinct subclass of halide-based materials, exhibiting superior chemo-mechanical stability alongside synthesis routes that appear more amenable to scale-up [31, 32]. These advances suggest promising prospects for the realization of low-cost, scalable oxyhalide solid electrolytes. Although there is still a lot to be explored for this new class of materials, they hold promise for application in industrially relevant solid-state batteries based on their ionic conductivity alone.

To illustrate the breadth of compositional space and performance variability in this materials class, Table 2 summarizes representative halide and oxyhalide solid electrolytes, highlighting their crystal structures, ionic conductivities, electrochemical windows, and known processing

or stability characteristics. These examples collectively demonstrate the wide range of tunability achievable through composition, structure, and anion substitution.

While this perspective focuses on halide solid electrolytes, it is worth briefly noting structural parallels to other emerging materials such as perovskite and antiperovskite electrolytes. Compounds like Li_3OCl share halide-rich frameworks with high ionic mobility and rely on mixed-anion sublattices to tune transport and stability—concepts that also underpin current efforts in oxyhalide development [34, 35]. Despite their conceptual relevance, these systems face major challenges with moisture stability and electrochemical compatibility, limiting their near-term application in practical batteries [33]. Nonetheless, they provide valuable structural and mechanistic insight that can inform the rational design of halide and mixed-anion solid electrolytes going forward.

These case studies—ranging from the well-established trigonal Li_3YCl_6 and spinel-type Li_2MgCl_4 to the mixed-anion oxyhalide LiNbOCl_4 —underscore the breadth of structural motifs accessible within the halide and halide-derivative design space. Each class presents distinct trade-offs, but collectively they illustrate the modularity of halide systems, where conductivity, interfacial behavior, and processability can be tuned through strategic choices in compositional design. As new halide chemistries continue to emerge, there is a growing need for coordinated efforts to map this complex design space with high-throughput experimentation, computational screening, and advanced characterization. The halide electrolyte family is no longer defined by a few benchmark materials—it is rapidly evolving into a chemically and structurally rich platform capable of meeting the diverse demands of next-generation solid-state batteries by exhibiting a unique blend of qualities that toe the line between the best qualities of oxides and the best qualities of sulfides.

Table 2 Representative halide and oxyhalide solid electrolytes: reported properties

Composition	Crystal structure	Ionic conductivity/ ($\text{mS}\cdot\text{cm}^{-1}$)	Stability window/(V vs. Li^+)	Moisture sensitivity	Doping strategy/notes	Refs.
Li_3YCl_6	Trigonal	$\sim 0.5\text{--}1.5$	$\sim 0.6\text{--}4.5$	Moderate	Al^{3+} , In^{3+} doping improves conductivity	[1]
Li_3InCl_6	Monoclinic	$\sim 0.4\text{--}1.0$	$\sim 0.5\text{--}4.2$	Moderate	Sc^{3+} doping improves ionic mobility	[2, 24]
Li_3ScCl_6	Monoclinic	$\sim 0.2\text{--}3$	$\sim 0.7\text{--}4.5$	Moderate	Narrower thermal stability window	[7]
Li_2ZrCl_6	Monoclinic	$\sim 0.1\text{--}0.7$	$\sim 1.5\text{--}4.2$	Moderate	Zr^{4+} alternative to rare-earth cations	[2, 26]
Li_2MgCl_4	Spinel	$\sim 1 \times 10^{-7}$	$\sim 1.2\text{--}3.8$	Unknown	Structural stability; conductivity enhanced with Mg doping for Li	[14, 27, 28]
Li_3YBr_6	Monoclinic	~ 1.0	$\sim 0.4\text{--}3.5$	Poor	Improved conductivity, lower oxidation stability	[1]
LiNbOCl_4	Tetragonal	$\sim 10\text{--}12$	$\sim 1.5\text{--}4.8$ (est.)	Unknown	Promising mixed-anion system	[5, 30, 33]
LiTaOCl_4	Tetragonal	~ 12	$\sim 1.5\text{--}5.0$ (est.)	Unknown	Potential air-stable halide derivative	[5, 30, 33]

4 Challenges

Despite their promising properties, halide solid electrolytes face critical challenges that must be addressed before they can be considered viable for commercial solid-state batteries. These challenges primarily center around materials stability, environmental compatibility, synthesis scalability, and long-term cell reliability.

4.1 Moisture sensitivity and environmental stability

Moisture sensitivity remains a central limitation in the practical deployment of halide solid electrolytes. While halides are generally more environmentally stable than sulfide electrolytes, most compositions—particularly chlorides and bromides—are still hygroscopic to varying degrees. Exposure to ambient humidity leads to hydrolysis reactions that form hydrated salts, oxychlorides, and acidic by-products such as HCl, which degrade ionic conductivity and can corrode manufacturing and processing equipment or cell components [8]. Even brief or partial exposure during powder handling, electrode mixing, or storage can result in irreversible changes to material performance, highlighting the importance of controlled-atmosphere processing and packaging.

The extent of hydrolysis in halide solid electrolyte materials depends on multiple factors, but primarily on the halide anion ($\text{Cl}^- > \text{Br}^- > \text{I}^-$ in terms of moisture reactivity) and the non-alkali cation. For example, the $\text{Li}_2\text{M}^{4+}\text{Cl}_6$ – $\text{Li}_3\text{M}^{3+}\text{Cl}_6$ system behaves differently depending on the transition-metal cation, with Li_3YCl_6 undergoing an irreversible hydration decomposition, Li_3InCl_6 undergoing a reversible hydration reaction, and Li_2ZrCl_6 irreversibly degrading through hydrolysis with HCl generation [8, 15, 36]. While spinel-type and oxyhalide systems may offer improved moisture tolerance due to more tightly packed lattice structures, lower surface energy, and lattice stabilization through halide substitution, these systems have yet to be thoroughly investigated for their moisture sensitivity enough to assess whether they are truly viable for gigafactory-scale material production and cell manufacturing. Regardless, halide solid electrolyte sensitivity to moisture can drastically affect the properties of the material as well as its interfacial stability and processability.

Beyond just the sensitivity of material performance to moisture exposure, the generation of HCl gas in some of the less expensive materials poses a serious problem. The evolution of HCl can cause the degradation of other cell components, such as the current collectors (e.g., Al or Cu), active materials, and binders, undermining mechanical adhesion and long-term electrochemical performance.

Furthermore, moisture sensitivity complicates efforts to scale halide electrolytes with conventional wet slurry processes for composite electrode fabrication, pushing development toward dry room compatible or fully solvent-free processing techniques.

Improving environmental stability will likely require a combination of materials design and systems-level strategies. At the materials level, compositional tuning through halide substitution or mixed-halide approaches, surface passivation via thin inorganic coatings, or nano-structuring to reduce reactive surface area are promising avenues [36]. At the device level, engineering solutions such as dry room assembly, hermetic packaging, and electrolyte encapsulation layers can help protect against incidental exposure. However, these measures add cost and complexity to material and cell manufacturing, underscoring the need for more intrinsically stable halide compositions.

While halides offer a safer alternative to sulfides by avoiding toxic gas release (e.g., H_2S), their environmental robustness still lags that of oxide-based electrolytes. Developing a deeper understanding of degradation mechanisms—particularly under realistic storage and cycling conditions—remains critical to ensuring halides can meet the shelf-life, safety, and performance requirements for commercial deployment.

4.2 Scalability of synthesis and processing

Translating halide solid electrolytes from laboratory materials to commercially viable battery components will require the development of synthesis routes that are not only chemically robust but also scalable, cost-effective, and compatible with industrial processing environments. While halides offer the advantage of relatively low-temperature and solvent-free synthesis compared to oxides, several challenges remain in ensuring batch-to-batch consistency, phase purity, and moisture protection at scale.

Mechanochemical milling has emerged as a widely used synthesis approach for halide electrolytes due to its solvent-free nature, low equipment requirements, and capacity for dopant incorporation. This technique has enabled rapid prototyping of various halide compositions; however, as production volumes increase, mechanochemical methods introduce variability in particle size distribution, local heating effects, and residual unreacted precursors—each of which can impact the degree of crystallinity, phase purity, and consequently, ionic conductivity and interfacial behavior. In-line process monitoring and rigorous precursor control will be required to mitigate these limitations in large-scale material production.

Solution-based approaches, such as co-precipitation or solvent-assisted mixing, offer improved stoichiometric control and may enable better dispersion of dopants [37].

However, the use of halide salts in aqueous or alcohol-based systems often necessitates extreme drying protocols and raises concerns over hydrolysis, halide volatilization, and environmental handling. These methods may be more suitable for pilot-scale batch processes than high-throughput continuous manufacturing.

Melt processing and quenching techniques—though less common—have shown promise for achieving a high degree of crystallinity and phase purity. These routes typically require inert atmospheres and high-purity precursors but offer the benefit of thermodynamic control over phase formation. Still, the energy and equipment demand of melt processing make it less attractive for cost-sensitive applications unless offset by performance gains or co-integration with cathode fabrication steps.

Each synthetic approach presents trade-offs in throughput, environmental control, and scalability. Table 3 summarizes the advantages and disadvantages of the most common synthesis routes for halide solid electrolytes explored in the literature [8, 14, 33, 38, 39]. An ideal route would combine the low-cost, solvent-free benefits of mechanochemistry with the compositional precision of solution methods, potentially through hybrid processes such as reactive extrusion or controlled-atmosphere ball milling. In parallel, there is a growing need to assess the lifecycle costs and environmental footprint of halide precursor supply chains—particularly for rare-earth-containing compositions like Li_3YCl_6 or Li_3InCl_6 —versus emerging alternatives based on earth-abundant elements like Mg, Fe, or Al.

Ultimately, the selection of a synthesis method will depend not only on electrolyte performance but also on its compatibility with downstream processing—such as composite cathode formation, film casting, or densification—and its resilience to ambient exposure during storage and assembly. Achieving this balance will be critical to enabling halide electrolytes to transition from the lab to a commercially viable solid-state electrolyte.

4.3 Long-term chemical and mechanical stability

Although halide solid electrolytes have shown promising electrochemical performance in initial tests, they are not chemically stable against lithium metal, and their long-term compatibility at the anode interface remains a critical concern. Upon contact with lithium, halides undergo spontaneous reduction reactions, forming decomposition products such as LiCl and mixed-metal halides [40]. These interphases, while sometimes passivating, are not inherently self-limiting and may continue to evolve during cycling. The ionic conductivity, mechanical adhesion, and electrochemical stability of these layers are poorly characterized and may vary widely depending on the halide composition, local current density, and cycling conditions. In practical cell configurations, the formation of resistive or non-uniform interphases can lead to increased interfacial impedance, uneven lithium plating, void formation, or even delamination of the anode–electrolyte interface. Thus, despite their relative improvement over sulfides in terms of oxidative stability, halides present a fundamentally different set of interfacial degradation risks when paired with lithium metal, and mitigation strategies must be incorporated into any viable full-cell architecture.

Mechanically, halide solid electrolytes tend to have a moderate elastic modulus ($\sim 10\text{--}30$ GPa), positioning them between brittle oxides and soft sulfides. This softness can promote better conformal contact at interfaces and mitigate interfacial stress during fabrication or early cycling. Recent work has shown that stack pressure plays a critical role in maintaining interface stability; when pressure is increased moderately (e.g., $2\text{--}14$ MPa), halide SSEs demonstrate superior cycling stability and capacity retention, but excess pressure (> 20 MPa) may provoke lithium metal creep and accelerate interfacial degradation. However, over-extended cycling, unless proper stack pressure is maintained, interfacial degradation may occur, especially in dry-processed or high-loading electrode stacks. Additionally, the potential for crack formation, GB sliding, or

Table 3 Summary of halide electrolyte synthesis routes

Synthesis method	Description	Advantages	Limitations
Mechanochemical milling	High-energy ball milling of halide precursors	Solvent-free, scalable, relatively low cost	Moisture sensitive, contamination risk, possible phase mixing
Melt quenching	Melting followed by rapid cooling	Good crystallinity, tunable phase control	Requires high-purity starting materials, high energy for high temperatures
Co-precipitation	Precipitation of halide salts from solution	Stoichiometry control, scalable for some systems	Residual solvent or water, hydrolysis during drying
Solid-state reaction	High-temperature annealing of mixed powders	Produces dense, high-purity phases	Energy-intensive, not ideal for large-scale synthesis
Solution combustion	Nitrate and halide precursors in an exothermic reaction	Fast, uniform synthesis, applicable to doped systems	Risk of incomplete combustion or residual unreacted products

intergranular void growth remains poorly characterized for this class of electrolyte materials and may become limiting factors in thick-electrode or multilayer pouch cell configurations, both of which are critical to realizing the desirable theoretical energy density of all-solid-state batteries [11, 13].

Halide SSEs also boast relatively high ionic conductivities and exceptional oxidative stability, enabling the use of high-voltage cathodes and supporting advanced cathode architectures [41, 42]. Recent materials-level advances, such as aliovalent doping in Li-Mg-Zr-Cl spinel-type electrolytes, have enhanced conductivity via engineered Li^+ vacancy content, opening new approaches for interface engineering and bulk phase stability. [14], Interface engineering strategies targeting the physical and chemical stability of SSEs, such as doping, interlayer construction, and scalable process optimization, are increasingly being adopted to suppress interfacial resistance, dendrite formation, and side reactions, especially as cells are cycled to high states of charge [43, 44].

Nonetheless, over long-term cycling and high-voltage operation, halide electrolytes are susceptible to gradual compositional drift, volatilization of surface-exposed halides, and oxidative side reactions at buried interfaces, the latter of which may evolve imperceptibly during many cycles despite the moderate operating voltages of most halide-compatible cathodes [42, 45]. Migration of halide ions or cation dopants under an electrochemical field may alter local conductivity or induce phase transformations, and failure to control GB environments can lead to capacity fading or increased impedance [11, 13]. Durable operation requires attention to both bulk and interface stability; recent studies with advanced synchrotron microscopy, X-ray photoelectron spectroscopy, impedance spectroscopy, and solid-state nuclear magnetic resonance spectroscopy have demonstrated the critical importance of meticulous stack pressure management and defect control for mitigating interfacial degradation mechanisms [44, 46].

Crucially, due to the inherent chemical instability of halides against lithium metal, engineered interlayers or artificial interphases are frequently employed to enable stable operation in lithium metal-based solid-state batteries [45]. These modifications must remain ionically conductive, chemically robust, and mechanically compliant throughout extended cycling, and ideally, they should be compatible with scalable manufacturing routes [43, 44]. The future development of halide-based SSBs will thus depend not only on the ongoing design of high-performing, defect-tolerant electrolyte compositions but also on the co-engineering of interfaces capable of sustaining multilayer pouch cell formats at high energy density without compromising safety or cycle life [13, 41].

5 Roadmap to commercialization

The path to commercializing halide-based SSBs is both promising and complex. Halide electrolytes enter the field at a time when the battery community is increasingly focused not only on performance metrics but also on manufacturability, scalability, and system-level integration. To gain traction in real-world applications, halide systems must offer not just good intrinsic material properties, but also provide a compelling combination of performance, ease of processing, and simplicity in scaling, or at least a significant benefit if scaling is complex.

In parallel with technical validation, cost competitiveness will be critical to the adoption of halide solid electrolytes. While some halides, such as Li_3YCl_6 and Li_3InCl_6 , exhibit excellent conductivity and cathode compatibility, their reliance on rare-earth elements like yttrium and indium raises concerns around raw material cost, sourcing risk, and long-term supply chain viability. In addition, the need for high-purity precursors and controlled synthesis conditions—such as inert atmosphere handling to prevent hydrolysis—can inflate production costs at scale. A major opportunity lies in identifying and optimizing alternative halide systems based on more abundant, lower-cost elements (e.g., Al, Mg, Fe) and developing scalable low-temperature synthesis approaches. A water-free solution process would likely be the easiest to scale. Moreover, if halide electrolytes can reduce the need for cathode coatings or enable simplified composite processing, these system-level cost offsets may compensate for their raw material premiums when compared to other solid electrolyte materials. Early techno-economic modeling tailored to halide systems is urgently needed to benchmark their potential cost-per-kWh against sulfide, oxide, and polymer-based SSBs. Figure 2 depicts a near, mid, and long-term development horizon outlining critical steps and requirements for the commercialization of halide solid electrolyte materials.

In the near term—over the next 1–3 years—the focus must be on establishing foundational protocols and reproducibility. This includes refining scalable synthesis techniques and standardizing the reporting of performance metrics such that they are reported consistently across studies to enable cross-laboratory comparisons. Demonstrations of stable long-term cell cycling with high-voltage or industrially relevant cathodes and thin lithium metal anodes will be a critical proof point. At the manufacturing level, the compatibility of halides with cold-press and solvent-free electrode processing should be leveraged to initiate pilot-scale composite cathode production. Early market applications may include niche sectors like Internet of Things devices or medical electronics, where moisture

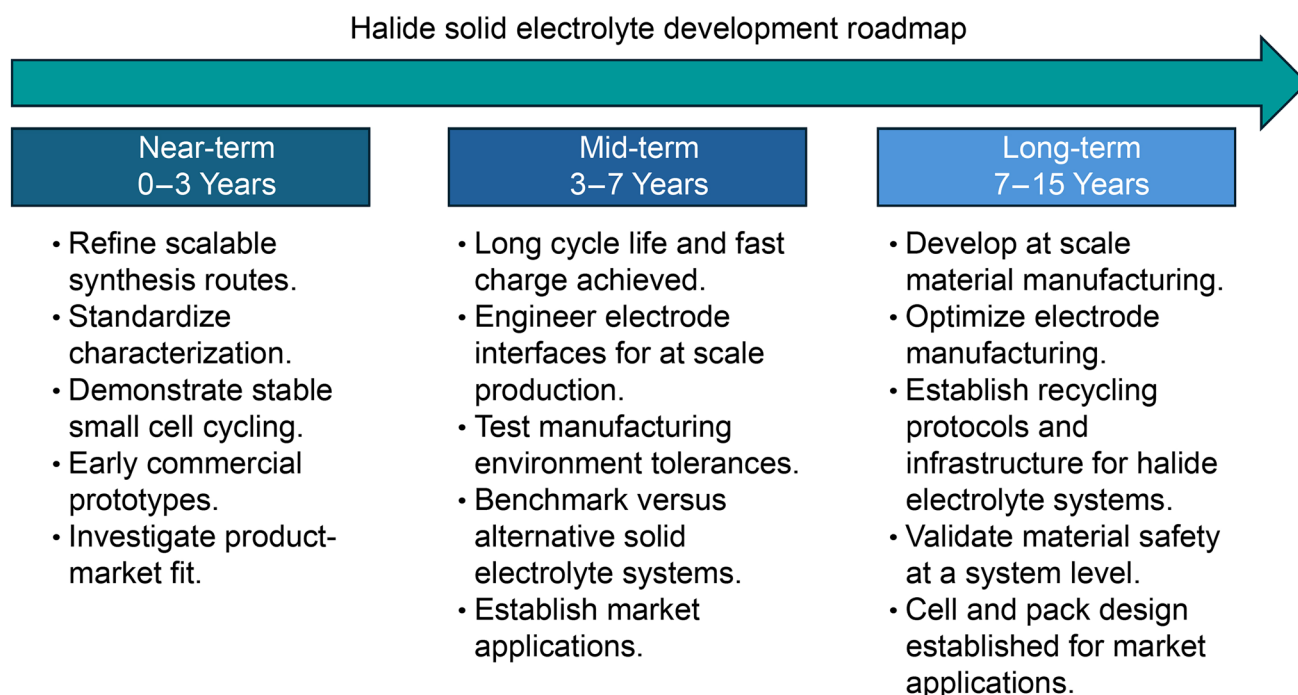


Fig. 2 Roadmap for R&D and commercial development of halide solid-state electrolyte materials and their corresponding all-solid-state batteries

sensitivity can be managed and the benefits of high energy density outweigh cost considerations.

In the medium term—3–7 years—the primary goal should shift toward cell integration and performance validation under realistic operating conditions for more mainstream applications. Target milestones will include achieving high-capacity retention (e.g., > 80% over 800+ cycles) in full cells with high areal loading (> 4 mAh·cm⁻²), operating at room temperature. This period will also require demonstrating stable operation at current densities above 1 mA·cm⁻², the development of engineered interfaces with lithium metal (with artificial solid electrolyte interphases or halide-based buffer layers), and improved moisture stability through innovations such as oxyhalides or surface treatments.

In the long term—beyond 7 years—halide-based batteries will need to prove themselves in mass-market applications such as electric vehicles, aerospace systems, and grid-scale installations. This will require not only performance parity with current state-of-the-art technologies but also system-level cost advantages and manufacturability. The development of dry, solvent-free, or water-free processing routes compatible with gigafactory-scale production will be a critical enabler. At the materials level, co-optimized halide–cathode chemistries may be needed to reach energy densities exceeding 800 Wh·L⁻¹ while maintaining safety and cycle life. Recycling and circular economy approaches for halide electrolytes will also become important depending on the prevailing halide electrolyte chemistry, as many of the

most heavily explored compositions have high-cost precursors with limited availability. Machine learning-guided discovery may accelerate the identification of next-generation halide or hybrid compositions with enhanced stability, ionic conductivity, and moisture tolerance [47, 48]. Meanwhile, rigorous safety validation, including thermal runaway tests and abuse tolerance assessments, is essential for commercial qualification.

Across all phases of development, several critical gaps remain. These include a lack of *operando* and *in situ* tools to observe halide behavior at interfaces and during cycling, a limited understanding of degradation pathways such as halogen migration or interfacial layer growth, and an absence of techno-economic modeling tailored to halide-based battery systems. To fulfill their potential, halide electrolytes must be pursued not as plug-and-play alternatives, but as a distinct materials class requiring new cell designs, interface engineering strategies, and process innovations. Their ultimate success will depend on coordinated, multidisciplinary efforts to bridge the gap from laboratory promise to commercial reality.

6 Conclusion and perspectives

Halide solid electrolytes have emerged as promising materials for next-generation solid-state batteries, offering a unique blend of high ionic conductivity, oxidative stability,

and processability. Their intermediate mechanical properties, compatibility with high-voltage cathodes, and potential for low-temperature, solvent-free processing position them between sulfide and oxide systems—leveraging the best qualities of both while avoiding many of their limitations.

Despite these advantages, halides face well-defined challenges that must be addressed for commercial adoption—particularly their chemical instability with lithium metal, sensitivity to moisture, and dependence on high-purity precursors. These limitations underscore the importance of robust interfacial engineering, scalable and dry room-compatible synthesis routes, and compositional strategies that prioritize both performance and manufacturability while not losing sight of long-term supply chain vulnerabilities. Yet because these challenges are tractable and known, halides may offer a faster path to manufacturable solid-state battery systems than more mature but increasingly constrained sulfide and oxide approaches.

While significant capital and infrastructure have been invested in sulfide and oxide electrolytes through companies like Solid Power and QuantumScape, respectively, halides remain an open field for a new entrant to make rapid progress. A well-positioned company that focuses early on scalable halide synthesis and reproducible lithium-metal interface solutions could avoid legacy limitations and move quickly toward commercially viable full-cell architectures. With the right priorities—dry room processing, interface co-design, and cost-sensitive materials selection—halide solid electrolytes could shift from promising alternatives to the foundation of practical, high-energy, solid-state batteries.

Acknowledgements The author would like to acknowledge Ieuan Seymour and Yutao Li for discussions that shaped the content of this perspective as well as Mojdeh Nikpour, Jack Kivi, and Hailong Lyu for their work on halide-related materials at EnergyX that led to some of the insights detailed in the present work.

Data availability No data was generated in the article.

Declarations

Conflict of interest The author has no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Asano T, Sakai A, Ouchi S, Sakaida M, Miyazaki A, Hasegawa S. Solid halide electrolytes with high lithium-ion conductivity for application in 4V class bulk-type all-solid-state batteries. *Adv Mater.* 2018;30(44):1803075. <https://doi.org/10.1002/adma.201803075>.
- Kim K, Park D, Jung HG, Chung KY, Shim JH, Wood BC, Yu, S. Material design strategy for halide solid electrolytes Li_3MX_6 (X = Cl, Br, and I) for all-solid-state high-voltage Li-ion batteries. *Chem Mater.* 2021;33(10):3669. <https://doi.org/10.1021/acs.chemmater.1c00555>.
- Luo M, Wang CH, Duan Y, Zhao XY, Wang JT, Sun XL. Surface coating enabling sulfide solid electrolytes with excellent air stability and lithium compatibility. *Energy Environ Mater.* 2024;7(6):e12753. <https://doi.org/10.1002/eem2.12753>.
- Li XN, Liang JW, Yang XF, Adair KR, Wang CH, Zhao FP, Sun XL. Progress and perspectives on halide lithium conductors for all-solid-state lithium batteries. *Energy Environ Sci.* 2020;13:1429. <https://doi.org/10.1039/C9EE03828K>.
- Tanaka Y, Ueno K, Mizuno K, Takeuchi K, Asano T, Sakai A. New oxyhalide solid electrolytes with high lithium ionic conductivity $>10 \text{ ms}\cdot\text{cm}^{-1}$ for all-solid-state batteries. *Angew Chem Int Ed.* 2023;62(13):e202217581. <https://doi.org/10.1002/anie.202217581>.
- Tuo K, Yin F, Sun C. Low-cost halide electrolytes $\text{Li}_{2+x}\text{Hf}_{1-x}\text{Fe}_x\text{Cl}_6$ with superior ionic conductivities for all-solid-state lithium-metal based batteries. *ACS Sustain Chem Eng.* 2024;12(18):7012. <https://doi.org/10.1021/acsschemeng.4c00365>.
- Liang J, Li X, Wang S, Adair KR, Li W, Zhao Y, Liang JW, Li XN, Wang S, Adair KR, Li WH, Zhao Y, Wang CH, Hu YF, Li Z, Zhao SQ, Lu SG, Huang H, Li RY, Mo YF, Sun XL. Site-occupation-tuned superionic $\text{Li}_x\text{ScCl}_{3+x}$ halide solid electrolytes for all-solid-state batteries. *J Am Chem Soc.* 2020;142(15):7012. <https://doi.org/10.1021/jacs.0c00134>.
- Chen YT, Tan DHS, Ham SY, Sayahpour B, Lee JB, Kim Y, Song MS, Nguyen LHB, Oh JAS, Ridley P, Cronk A, Deysher G, Jang J, Chen Z, Meng YS. Investigating dry room compatibility of chloride solid-state electrolytes for scalable manufacturing. *J Electrochem Soc.* 2023;170(8):080521. <https://doi.org/10.1149/1945-7111/acee24>.
- Wu F, Fitzhugh W, Ye L, Ning J, Li X. Advanced sulfide solid electrolyte by core-shell structural design. *Nat Commun.* 2018;9(1):4037. <https://doi.org/10.1038/s41467-018-06123-2>.
- Liang JW, Maas EV, Luo J, Li XN, Chen N, Adair KR, Li WH, Li JJ, Hu YF, Liu J, Zhang L, Zhao SQ, Lu SG, Wang JT, Huang H, Zhao WX, Parnell S, Smith RI, Ganapathy S, Wagemaker M, Sun XL. A series of ternary metal chloride superionic conductors for high-performance all-solid-state lithium batteries. *Adv Energy Mater.* 2022;12(21):2103921. <https://doi.org/10.1002/aenm.202103921>.
- Tao B, Zhong D, Li H, Wang G, Chang H. Halide solid-state electrolytes for all-solid-state batteries: structural design, synthesis, environmental stability, interface optimization and challenges. *Chem Sci.* 2023;14:8693. <https://doi.org/10.1039/D3SC02093B>.
- Kwak H, Wang S, Park J, Liu YS, Kim KT, Choi Y, Mo YF, Jung YS. Emerging halide superionic conductors for all-solid-state batteries: design, synthesis, and practical applications. *ACS Energy Lett.* 2022;7(5):1776. <https://doi.org/10.1021/acsenergylett.2c00438>.
- Li C, Du Y. Building a better all-solid-state lithium-ion battery with halide solid-state electrolyte. *ACS Nano.* 2025;19(4):4121. <https://doi.org/10.1021/acsnano.4c15005>.
- Rom CL, Yox P, Cardoza AM, Smaha RW, Phan MQ, Martin TR, Maughan AE. Expanding the phase space for

- halide-based solid electrolytes: Li–Mg–Zr–Cl spinels. *Chem Mater.* 2024;36(15):7283. <https://doi.org/10.1021/acs.chemmater.4c01160>.
15. Wang K, Gu ZQ, Liu HX, Lv Hu, Wu Y, Xu J, Ma C. High-humidity-tolerant chloride solid-state electrolyte for all-solid-state lithium batteries. *Adv Sci.* 2024;11(14):2305394. <https://doi.org/10.1002/adv.202305394>.
 16. Subramanian Y, Rajagopal R, Ryu KS. Toward achieving a high ionic conducting halide solid electrolyte through low-cost metal (Zr and Fe) and F substitution and their admirable performance in all-solid-state batteries. *ACS Appl Mater Interfaces.* 2024;16(19):24534. <https://doi.org/10.1021/acsami.4c01352>.
 17. Momma K, Izumi F. Vesta 3 for three-dimensional visualization of crystal, volumetric and morphology data. *J Appl Crystallogr.* 2011;44(6):1272. <https://doi.org/10.1107/S0021889811038970>.
 18. Yang S, Kim SY, Chen G. Halide superionic conductors for all-solid-state batteries: effects of synthesis and composition on lithium-ion conductivity. *ACS Energy Lett.* 2024;9(5):2212. <https://doi.org/10.1021/acsenergylett.4c00317>.
 19. Wang QD, Zhou YN, Wang XL, Guo H, Gong SP, Yao ZP, Wu FT, Wang JL, Ganapathy S, Bai XD, Li BH, Zhao CL, Janek J, Wagemaker M. Designing lithium halide solid electrolytes. *Nat Commun.* 2024;15(1):1050. <https://doi.org/10.1038/s41467-024-45258-3>.
 20. Nie X, Hu J, Li C. Halide-based solid electrolytes: the history, progress, and challenges. *Interdiscip Mater.* 2023;2(3):365. <https://doi.org/10.1002/idm2.12090>.
 21. Flores-González N, Minafra N, Dewald G, Reardon H, Smith RI, Adams S, Zeier WG, Gregory DH. Mechanochemical synthesis and structure of lithium tetrahaloaluminates, LiAlX_4 (X = Cl, Br, I): a family of Li-ion conducting ternary halides. *ACS Mater Lett.* 2021;3(5):652. <https://doi.org/10.1021/acsmaterialslett.1c00055>.
 22. Barker K, McKinney SL, Artal R, Jiménez R, Tapia-Ruiz N, Skinner SJ, Aguadero A, Seymour ID. The importance of a-site cation chemistry in superionic halide solid electrolytes. *Nat Commun.* 2024;15(1):7501. <https://doi.org/10.1038/s41467-024-51710-1>.
 23. Wang C, Liang J, Kim JT, Sun X. Prospects of halide-based all-solid-state batteries: from material design to practical application. *Sci Adv.* 2022;8(36):eadc9516. <https://doi.org/10.1126/sciadv.adc9516>.
 24. Li XN, Liang JW, Luo J, Banis MN, Wang CH, Li WH, Deng SX, Yu C, Zhao FP, Hu YF, Sham TK, Zhang L, Zhao SQ, Lu SG, Huang H, Li RY, Adair KR, Sun XL. Air-stable Li_3InCl_6 electrolyte with high voltage compatibility for all-solid-state batteries. *Energy Environ Sci.* 2019;12:2665. <https://doi.org/10.1039/C9EE02311A>.
 25. Miyazaki RM, Ozaki S, Yagi E, Sato Y, Kobayashi Y, Yoshida T, Katsuda Y. All-solid-state lithium-ion batteries using the Li_3AlF_6 -based composite as the solid electrolyte. *ACS Appl Energy Mater.* 2022;5(12):15365. <https://doi.org/10.1021/acsaem.2c02985>.
 26. Shi JT, Yao ZJ, Xiang JY, Cai C, Tu FF, Zhang YQ, Yao WL, Jia QX, Zhou Y, Shen SH, Yang YF. High-conductivity Li_2ZrCl_6 electrolytes via an optimized two-step ball-milling method for all-solid-state lithium–metal batteries. *ACS Sustain Chem Eng.* 2024;12(5):2009. <https://doi.org/10.1021/acssuschemeng.3c06652>.
 27. Lutz HD, Schmidt W, Haeuseler H. Chloride spinels: a new group of solid lithium electrolytes. *J Phys Chem Solids.* 1981;42(4):287. [https://doi.org/10.1016/0022-3697\(81\)90142-6](https://doi.org/10.1016/0022-3697(81)90142-6).
 28. Gao Z, Wang SW, Sun F, Yu ZQ, Song HC, Liu ZQ, Chen HJ. Enhanced electrochemical properties of Li_2MgCl_4 by Zn substitution for all-solid-state batteries. *J Solid State Chem.* 2023;328:124361. <https://doi.org/10.1016/j.jssc.2023.124361>.
 29. Zhou L, Kwok CY, Shyamsunder A, Zhang Q, Wu X, Nazar LF. A new halospinel superionic conductor for high-voltage all solid state lithium batteries. *Energy Environ Sci.* 2020;13:2056. <https://doi.org/10.1039/D0EE01017K>.
 30. Adams S. Origin of fast Li^+ -ion conductivity in the compressible oxyhalide LiNbOCl_4 . *Energy Storage Mater.* 2024;68:103359. <https://doi.org/10.1016/j.ensm.2024.103359>.
 31. Wang GZ, Zhang SM, Wu H, Zheng M, Zhao CT, Liang JW, Zhou LY, Yue JY, Zhu XZ, Xu Y, Zhang N, Pang TL, Fu JM, Li WH, Xia YG, Yin W, Sun XL, Li XN. Oxychloride polyanion clustered solid-state electrolytes via hydrate-assisted synthesis for all-solid-state batteries. *Adv Mater.* 2025;37(4):2410402. <https://doi.org/10.1002/adma.202410402>.
 32. Dai T, Wu SY, Lu YX, Yang Y, Liu Y, Chang C, Rong XH, Xiao RJ, Zhao JM, Liu YH, Wang WH, Chen LQ, Hu YS. Inorganic glass electrolytes with polymer-like viscoelasticity. *Nat Energy.* 2023;8(11):1221. <https://doi.org/10.1038/s41560-023-01356-y>.
 33. Wang CH, Liang JW, Luo J, Liu J, Li XN, Zhao FP, Li RY, Huang H, Zhao SQ, Zhang L, Wang JT, Sun XL. A universal wet-chemistry synthesis of solid-state halide electrolytes for all-solid-state lithium-metal batteries. *Sci Adv.* 2021;7(37):eabh1896. <https://doi.org/10.1126/sciadv.abh1896>.
 34. Zhao Y, Daemen LL. Superionic conductivity in lithium-rich antiperovskites. *J Am Chem Soc.* 2012;134(36):15042. <https://doi.org/10.1021/ja305709z>.
 35. Xia W, Zhao Y, Zhao FP, Adair K, Zhao R, Li S, Zou RQ, Zhao YS, Sun XL. Antiperovskite electrolytes for solid-state batteries. *Chem Rev.* 2022;122(3):3763. <https://doi.org/10.1021/acs.chemrev.1c00594>.
 36. Jia QX, Yao ZJ, Xiang JY, Shi JT, Zhou Y, Huang JH, Zhang HL, Zhang XX, Yang YF, Tu JP. Rare earth metal ion-doped halide solid electrolytes plus Ta^{5+} substitution for long cycling all-solid-state batteries. *Adv Funct Mater.* 2025;35(22):2426053. <https://doi.org/10.1002/adfm.202426053>.
 37. Li XN, Liang JW, Chen N, Luo J, Adair KR, Wang CC, Banis MN, Sham TK, Zhang L, Zhao SQ, Lu SG, Huang H, Li RY, Sun XL. Water-mediated synthesis of a superionic halide solid electrolyte. *Angew Chem Int Ed.* 2019;131(46):16579. <https://doi.org/10.1002/ange.201909805>.
 38. Kato Y, Hori S, Saito T, Suzuki K, Hirayama M, Mitsui A, Yone-mura M, Iba H, Kanno R. High-power all-solid-state batteries using sulfide superionic conductors. *Nat Energy.* 2016;1(4):1. <https://doi.org/10.1038/nenergy.2016.30>.
 39. Xiao Y, Wang Y, Bo SH, Kim JC, Miara LJ, Ceder G. Understanding interface stability in solid-state batteries. *Nat Rev Mater.* 2020;5(2):105. <https://doi.org/10.1038/s41578-019-0157-5>.
 40. Kim W, Noh J, Lee S, Yoon K, Han S, Yu S, Ko KH, Kang K. Aging property of halide solid electrolyte at the cathode interface. *Adv Mater.* 2023;35(32):2301631. <https://doi.org/10.1002/adma.202301631>.
 41. Wu J, Li J, Yao X. Exploring the potential of halide electrolytes for next-generation all-solid-state lithium batteries. *Adv Funct Mater.* 2025;35(10):2416671. <https://doi.org/10.1002/adfm.202416671>.
 42. Song ZY, Wang TR, Yang H, Kan WH, Chen YW, Yu Q, Wang LK, Zhang YN, Dai YM, Chen HC, Yin W, Honda T, Avdeev M, Xu HH, Ma JW, Huang YH, Luo W. Promoting high-voltage stability through local lattice distortion of halide solid electrolytes. *Nat Commun.* 2024;15(1):1481. <https://doi.org/10.1038/s41467-024-45864-1>.
 43. Zhao P, Kainat J, Lan J, Yu Y, Lee YS, Yang X. Air stability of inorganic electrolyte for all-solid-state lithium batteries: advances, challenges, and perspectives. *Small.* 2025;21(32):2504753. <https://doi.org/10.1002/smll.202504753>.
 44. He HG, Wang LT, Al-Abbasi M, Cao CY, Li H, Xu Z, Chen S, Zhang W, L RQ i, Lai YK, Tang YX, Ge MZ. Interface engineering on constructing physical and chemical stable solid-state

- electrolyte toward practical lithium batteries. *Energy Environ Mater*. 2024;7(4):e12699. <https://doi.org/10.1002/eem2.12699>.
45. Ganesan P, Soans M, Cambaz MA, Zimmermanns R, Gond R, Fuchs S, Y Hu, Baumgart S, M Sotoudeh, Stepien D, Stein H, Groß A, Bresser D, Varzi A, Fichtner M. Fluorine-substituted halide solid electrolytes with enhanced stability toward the lithium metal. *ACS Appl Mater Interfaces*. 2023;15(32):38391. <https://doi.org/10.1021/acsami.3c03513>.
46. Kwak H, Kim JS, Han D, Kim JS, Park J, Kwon G, Bak SM, Heo U, Park C, Lee HW, Nam KW, Seo DH, Jung YS. Boosting the interfacial superionic conduction of halide solid electrolytes for all-solid-state batteries. *Nat Commun*. 2023;14(1):2459. <https://doi.org/10.1038/s41467-023-38037-z>.
47. Sendek AD, Cubuk ED, Antoniuk ER, Cheon G, Cui Y, Reed EJ. Machine learning-assisted discovery of solid Li-ion conducting materials. *Chem Mater*. 2019;31(2):342. <https://doi.org/10.1021/acs.chemmater.8b03272>.
48. Grundish NS, Ransom B, Sendek AD, Pellouchoud LA, Li Y, Reed EJ. Predicting anion redox in secondary battery cathode materials with a data-driven model. *J Phys Chem C*. 2024;128(40):16844. <https://doi.org/10.1021/acs.jpcc.4c02079>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.